Tutorial 6

Getting started with SPSS or PSPP

An easy video tutorial on:

SPSS https://www.youtube.com/watch?v=ZpwZS3XnEZA

PSPP https://www.youtube.com/watch?v=NqS6DTXeXC4

Activity 1

RODENTS IN SPACE: A SIMPLE MEMORY EXPERIMENT

Twenty one first-year undergraduates participated in a simple memory experiment designed to investigate the effect of a mnemonic strategy upon memory for paired words. The participants were randomly divided into two groups. All participants were given 2 minutes to memorise a list of 20 words presented in pairs. After learning the words for 2 minutes the participants were then required to recall any of the words that they could remember. The groups were split into a mnemonic group (participants were advised to try to form a mental image to link to the words) and a non-mnemonic group. The data are given below:

Memory scores (out of 20) for the mnemonic group:

20, 18, 14, 18, 17, 11, 20, 18, 20, 19, 20

Memory scores (out of 20) for the non-mnemonic group:

10, 20, 12, 9, 14, 15, 16, 14, 19, 12

Using these data to attempt the following:

1. Form a research question, which examines the difference between the two groups.
   RQ: Does the use of mnemonics affect memory?
2. Form a hypothesis, which examines the difference between the two groups.
   RQ: Does the use of mnemonics affect memory?
3. What is the dependent variable?
4. What is the independent variable?
5. In SPSS or PSPP, set up a data file to record the above data. More specifically:

- In **Variable View** (where we set up the independent variable and dependent variable):
    - In Name, give an appropriate Name to each variable.
    - In Type, insert Numeric for both variables
    - In Width, give an appropriate width to both variables.
    - In Decimals, type 0 for both variables.
    - In Label, give appropriate label to each variable.
    - In Values, apply a label and value to the variable which is responsible for splitting the sample.
    - In Measure, select Scale (which is either interval or ratio). Avoid using a nominal or ordinal variable as a dependent variable.
    - Regarding the rest variable properties, leave them as they are.
- **In Data View** (where we set up the observations for each variable), insert the above observations (data) in the column of the appropriate variable.
    - Save the .sav file, as 'MemoryStudy'.
    - Select Analyze > Descriptive Statistics > Frequencies, select both variables, select Statistics and select all the following options: Mean, Median, Mode, Std. deviation, Variance, Range, Minimum, Maximum, Skewness and Kurtosis. In Charts select Histograms and show normal curve and histogram. Look at the generated table, consult your lecture notes and discuss the output.
    - Answer the following questions:
        - Do we split the sample in this type of study according to treatment?
        - Is this type of study a 'between-subjects design' or a 'within-subjects design'? To answer the last question, perform research using 'Google' regarding the terms: 'between-subjects design' and 'within-subjects design'.

Activity 2

We conducted a study with 19 participants to test two different prototypes for the website https://www.snapfish.co.uk/photo-books where you can create and order a photo-book. Participants were asked to perform a number of tasks. One of the tasks required participants to upload 4 photos and then record the time (in seconds) in Prototype A (Pr.A), and then do the same for Prototype B (Pr.B). The data collected is provided below:

Participant/Case	Time of task in Pr.A	Time of task in Pr.B
1	30	65
2	60	140
3	80	135
4	120	180
5	50	145
6	31	90
7	55	180
8	33	42
9	75	54
10	58	60
11	47	25
12	80	45
13	70	52
14	170	28
15	88	78
16	170	42
17	150	65
18	44	21
19	32	40

Attempt the following:

1. Form a research question that examines the difference in terms of time for Prototype A and Prototype B.
2. Form a hypothesis, which examines the difference between the two groups.
3. What is the dependent variable?
4. What is the independent variable?

5. In SPSS or PSPP, set up a data file to record the above data. More specifically:
   - In **Variable View**, apply the appropriate Variable properties for the independent and dependent variables. (Tip: Look at how you did this the previous exercise).
   - In **Data view**, insert the above observations/data.
   - Save the .sav file, as 'PhotoBookStudy'.

6. Answer the following questions:
   a. Do we split the sample in this type of study according to treatment?
   b. Is this type of study a 'between-subjects design' or a 'within-subjects design'? To answer the last question, perform a research using 'Google' regarding the terms: 'between-subjects design' and 'within-subjects design'.

7. Use SPSS or PSPP to find the mean regarding the time for each prototype.
   a. Go Analyse>Descriptive Statistics>Explore
   b. In the Dependent List select the upload time, as factor list select the prototype used. In Statistics select Descriptives with Confidence Interval for Mean 95%. In plots select both Steam-and-leaf and Histogram. Select normality plots with tests. Look at the generated table and graphs, consult your lecture notes and discuss the output.
   c. Save it in the same folder as the PhotoBookStudy.sav (and with the same name).

   Think: After, finding the mean statistic, is there a significant difference between the two means? Can we really judge how significant –how big- is their difference based on Descriptive statistics (i.e. calculation of the two means)? Can we really answer the research question based only on the calculation of the means?

Activity 3

Read and reflect on an interesting article about mean and mode

Mean is the most popular way to calculate the average, even when it is not appropriate to use it (e.g. when calculating the average of ordinal variables such as 'number of children a woman has in UK'). However, read an article where mode gives us more interesting and valuable information: 'Small Data: When average age of death was zero' (source: http://www.bbc.co.uk/news/blogs-magazine-monitor-29618190)